TriFrame: can pictures of stock charts help predict risk?

Computer Vision with Deep Learning project · end-to-end report · 120-stock run (s120, notebook v3) · generated 2026-10-20

In one paragraph. We asked whether pictures of a stock's price chart at three time scales (weekly, daily and hourly) help a deep-learning model predict three things about the coming weeks: how much the price will swing (**volatility**), whether it will fall 10% or more at some point (**drawdown**), and whether it will beat the typical stock over the next 5 days (**direction**). We tested 7 pretrained vision models (each both frozen and fine-tuned) on 120 US stocks (56,099 chart sets, 2001-2022), always against a simple baseline that uses only 33 numbers computed from the same prices. **The answer:** risk is predictable (volatility R^2 0.50 on the unseen 2020-2022 test years), but the chart pictures did not add a significant improvement over the numbers there, and there is no reliable 5-day direction signal, so the system's default decision is **NO TRADE**.

Target	Model chosen	Test score (2020-2022)	Sealed holdout (2023-2026)	In plain words
Volatility, next 20 days	ViT-B/16 (fine-tuned), pretrained on ImageNet-21k + 1k + gradient-boosted trees (XGBoost)	R^2 0.50	R^2 0.58	explains about 50% of the variation in volatility
10%+ drawdown within 20 days	numbers only + gradient-boosted trees (XGBoost)	AUC 0.66	not opened	ranks a stock that drops 10% above one that does not 66% of the time
Beats the median stock, 5 days	numbers only + gradient-boosted trees (XGBoost)	AUC 0.508	AUC 0.508	no better than a coin flip

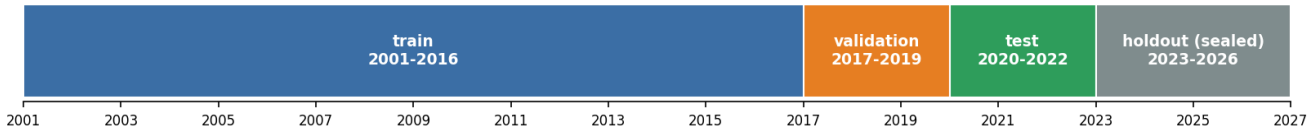
What is new in this version (v3)

- Trained on **120 stocks** instead of 60 (the new sample contains the pilot's 60, so the two runs compare like for like).
- Every pretrained model is **fine-tuned** on the charts as well as used frozen, and both compete under the same rule.
- A **leaderboard** scores every model on six metrics per target, with confidence intervals and charts.
- A **Monte Carlo simulation** of the next 20 trading days, calibrated on validation data and checked on the test years.
- The dashboard analyses **up to four companies at once**, ranks them by risk and explains every number in plain words.

Steps 1-2: the data and how it was split

Each example, a **chart set**, is one stock on one day: three black-and-white chart pictures drawn from its price history (104 weekly bars, 126 daily bars and 140 hourly bars) plus 33 ordinary numbers (returns, price ranges, volatility measures) computed from the last 126 daily bars. The labels are what happened next: the realised volatility over the next 20 trading days, whether the price traded 10% below the next day's open within 20 days, and whether the stock beat the median stock over 5 days. This run uses **120 of 979 US stocks** (chosen at random, and including the 60 stocks of the earlier pilot), 56,099 chart sets in total.

Time is never shuffled. Models learn from the **training** years, are tuned and selected on the **validation** years, and are scored once on the **test** years. 2023 onward is a sealed **holdout** that can be opened only once, by the owner, after everything is frozen. A 20-day gap at each boundary stops any label from leaking into the next period.



Period	Years	Chart sets	Stocks	Used for
train	2001-2016	38,937	117	fit the models
validation	2017-2019	8,496	118	tune, select, calibrate, set the NO TRADE rule and the simulation
test	2020-2022	8,666	120	one honest check of the frozen choices
holdout	2023-2026	-	-	sealed; opened once by the owner (Stage 10) - opened



What a model sees: the three pictures of one chart set (drawn here from synthetic prices). Each picture is scaled to fill its panel, so it shows the shape of the moves, not their size in dollars.

Step 3: the pretrained vision models

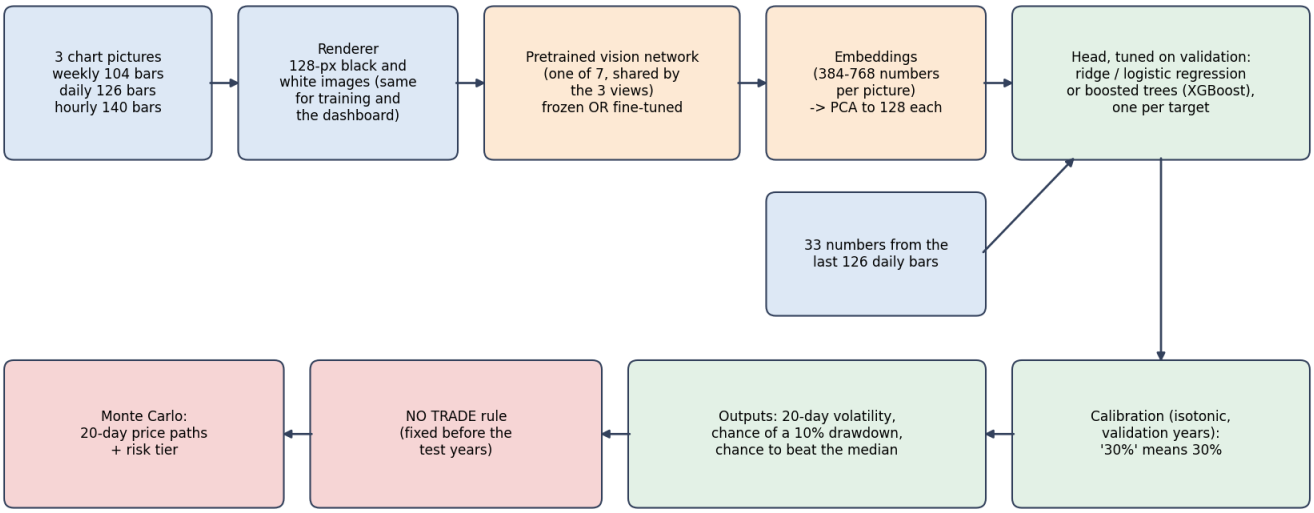
A **pretrained** model is a neural network that has already learned to recognise edges, shapes and textures from millions of ordinary photos. We reuse that knowledge instead of training a network from scratch. We compared seven well-known designs: convolutional networks (CNNs), which scan the picture with small filters, and transformers, which split it into patches that 'pay attention' to each other. Three ResNet-18s with **random, untrained weights** serve as a control: if they did as well, pretraining would not be doing anything.

Model	Type	Pretrained on	Parameters (millions)	Fine-tuning
ResNet-18	CNN	ImageNet-1k (1.28 million labelled photos, 1,000 classes)	11.2	3 epochs (best 1), 2 min
EfficientNet-B0	CNN	ImageNet-1k (1.28 million labelled photos, 1,000 classes)	4.0	3 epochs (best 1), 5 min
ViT-B/16	Transformer	ImageNet-21k, then fine-tuned on ImageNet-1k	85.8	4 epochs (best 2), 8 min
ConvNeXt-Tiny	CNN	ImageNet-12k, then fine-tuned on ImageNet-1k	27.8	4 epochs (best 2), 6 min
Swin-T	Transformer	ImageNet-1k (1.28 million labelled photos, 1,000 classes)	27.5	4 epochs (best 3), 7 min
DINOv2 ViT-S/14	Transformer	LVD-142M, 142 million unlabelled images (self-supervised)	21.6	4 epochs (best 3), 5 min
MobileNetV3-Large	CNN	ImageNet-1k (1.28 million labelled photos, 1,000 classes)	4.2	4 epochs (best 2), 5 min
ResNet-18 (random weights, seed 0)	Control	none (random weights)	11.2	-

Model	Type	Pretrained on	Parameters (millions)	Fine-tuning
ResNet-18 (random weights, seed 1)	Control	none (random weights)	11.2	-
ResNet-18 (random weights, seed 2)	Control	none (random weights)	11.2	-

Frozen or fine-tuned? 'Frozen' means the network's own weights are left exactly as they came from pretraining; we only read its output (an **embedding**, a list of a few hundred numbers summarising the picture) and train small models on top. 'Fine-tuned' means the network's weights are also adjusted on our charts. **In this run all 7 pretrained networks were fine-tuned** (up to 4 epochs on the training years, early stopping on validation) and competed with their frozen versions.

Step 4: the architecture



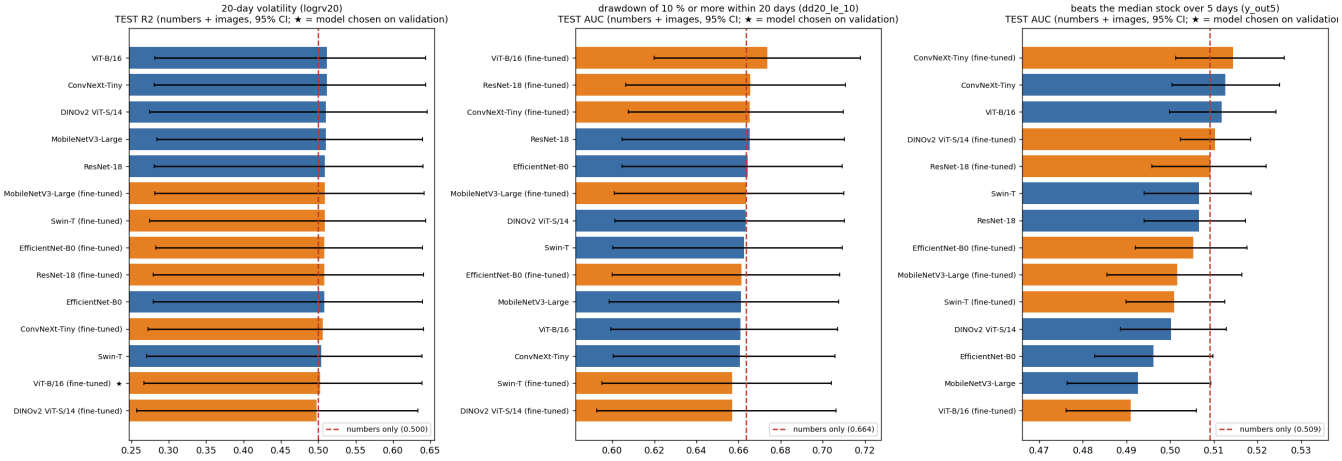
- The renderer draws the three charts as 128-pixel-high black-and-white pictures; the dashboard uses the very same code.
- One vision network turns each picture into an embedding; **PCA** compresses each embedding to 128 numbers (fitted on training years only).
- The three compressed embeddings and the 33 numbers go into a **head**: ridge or logistic regression, or gradient-boosted decision trees (XGBoost). One head per target.
- Probabilities are **calibrated** on the validation years so that '30%' really means 30%.
- A **NO TRADE rule**, fixed before the test years were opened, turns the outputs into a decision, and a **Monte Carlo simulation** turns the volatility forecast into a fan of possible price paths.

Step 5: training and hyper-parameter tuning

We tried **1,690 model configurations** and logged every one: ridge strengths, logistic regularisation, tree depth, learning rate and the number of trees (by early stopping), for each target, each vision network and each combination of inputs (images only, numbers + all three timeframes, numbers + one timeframe). Every choice was made on the validation years only, and all preprocessing (gap filling, scaling, PCA) was fitted on the training years only. **The selection rule was written down before the test years were opened:** per target the best validation score wins, and a model that uses chart pictures wins only if its improvement over the numbers-only model is clearly above zero (its 95% confidence interval excludes 0).

Step 6: results - which models won, and how good they are

Target	Inputs	Vision network	Head	Validation	Test (95% CI)
Volatility, next 20 days	33 numbers + weekly + daily + hourly charts	ViT-B/16 (fine-tuned), pretrained on ImageNet-21k + 1k	gradient-boosted trees (XGBoost)	R^2 0.583	R^2 0.503 (0.27 to 0.64)
10%+ drawdown within 20 days	33 numbers from the last 126 daily bars	none (numbers only)	gradient-boosted trees (XGBoost)	AUC 0.713	AUC 0.663 (0.60 to 0.71)
Beats the median stock, 5 days	33 numbers from the last 126 daily bars	none (numbers only)	gradient-boosted trees (XGBoost)	AUC 0.522	AUC 0.508 (0.496 to 0.520)



Every candidate on the test years (numbers + images, 95% CIs, ★ = the model chosen on validation). Descriptive only: the choice was frozen before these numbers were computed.

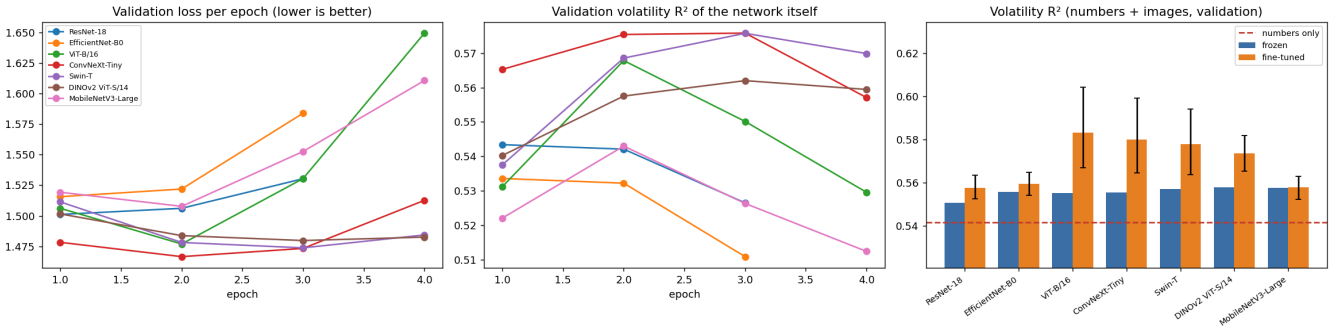
- **Do chart pictures add to the numbers? No.** On the test years, adding all three chart views to the numbers changed volatility R^2 by +0.003 (95% CI -0.016 to +0.027) and direction AUC by +0.000 (95% CI -0.014 to +0.015). Both intervals include zero.
- **Do the vision models differ from each other? Yes, a little.** 13 of 21 pairs of frozen pretrained networks differ in how well their pictures alone predict volatility (validation, Holm-corrected); the best reaches R^2 0.245.
- **Does waiting for confident direction calls help? No.** Keeping only the 10% most confident direction calls changed their accuracy by +0.004 (95% CI -0.019 to +0.029) on the test years.
- **Sanity checks.** With shuffled labels the pipeline finds nothing (volatility R^2 -0.107, direction AUC 0.504), and randomly initialised ResNet-18s reach volatility R^2 -0.001 to 0.005 from pictures alone, against 0.245 for the best pretrained network.

Step 7: did we fine-tune, and did it help?

Fine-tuning raised validation volatility R^2 (numbers + images) for 7 of 7 backbones (significantly for 5, significantly worse for 0; mean change +0.0142) and drawdown AUC for 6 of 7 (significantly for 1). Images-only volatility R^2 moved from 0.192 (frozen, average) to 0.215 (fine-tuned). Total fine-tuning time 38 GPU-minutes.

Model	Frozen R^2	Fine-tuned R^2	Change (95% CI)	Training time
ResNet-18	0.551	0.558	+0.007 (+0.002 to +0.013)	2 min
EfficientNet-B0	0.556	0.559	+0.004 (-0.002 to +0.009)	5 min
ViT-B/16	0.555	0.583	+0.028 (+0.012 to +0.049)	8 min

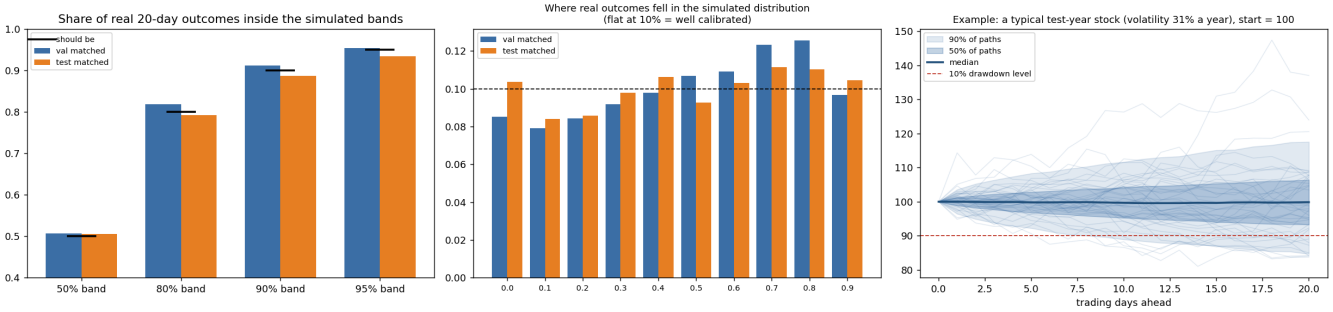
Model	Frozen R²	Fine-tuned R²	Change (95% CI)	Training time
ConvNeXt-Tiny	0.556	0.580	+0.024 (+0.009 to +0.044)	6 min
Swin-T	0.557	0.578	+0.021 (+0.007 to +0.037)	7 min
DINOv2 ViT-S/14	0.558	0.574	+0.016 (+0.008 to +0.024)	5 min
MobileNetV3-Large	0.558	0.558	+0.000 (-0.005 to +0.005)	5 min



Step 8: from predictions to a decision, and the Monte Carlo simulation

NO TRADE rule. A stock is flagged as high risk when its predicted volatility is at least 34% a year or its drawdown chance at least 28% (the top 20% of the model's readings in the validation years). On the test years the flag picked out 50% of cases, and those cases really were riskier: realised volatility 54% vs 28% a year, and 10% drawdowns 40% vs 19% of the time. A direction call would only ever be shown if the direction model had passed its validation *and* test checks; it did not, so every decision is NO TRADE.

Monte Carlo. For each stock the dashboard simulates thousands of 20-day price paths. Each path draws a volatility from the model's forecast and its measured error on validation data, then fat-tailed daily moves (Student-t with 5 degrees of freedom, chosen on validation) with exactly that volatility; daily lows reach below the closes by a range also fitted on validation. There is no drift, because the direction model has no edge. By default the paths are reweighted, as little as possible and without changing the average return, so the share that touches a 10% drawdown equals the calibrated drawdown model's chance. **Check on the test years:** the simulation's 90% band contained 89% of the real 20-day outcomes (validation: 91%) and its 50% band 51%.



Step 9: the dashboard - which models it uses

The *Computer Vision with Deep Learning Project Dashboard* runs the frozen models from the private model repository. You choose 1 to 4 companies and upload three TradingView screenshots (1W, 1D, 1H) for each. The app reads the ticker and the last bar of every screenshot, confirms them against market data, rebuilds exactly the inputs the models were trained on, and shows the outputs with charts and a plain-language explanation, a risk tier, a Monte Carlo fan of price paths and, for several companies, a ranking from riskiest to safest.

Output	Inputs	Vision network	Head	Test score
Volatility over the next 20 trading days	33 numbers + weekly + daily + hourly charts	ViT-B/16 (fine-tuned), pretrained on ImageNet-21k + 1k	gradient-boosted trees (XGBoost)	R^2 0.50
Chance of a 10%+ drawdown within 20 trading days	33 numbers from the last 126 daily bars	none (numbers only)	gradient-boosted trees (XGBoost)	AUC 0.66
Chance of beating the median stock over 5 trading days	33 numbers from the last 126 daily bars	none (numbers only)	gradient-boosted trees (XGBoost)	AUC 0.508

Does more data help? 120 stocks vs the 60-stock pilot

The pilot trained on 60 stocks; this run trained on 120, including those 60. Scoring both frozen models on the pilot's 60 stocks and the same test rows isolates the effect of more training data.

Target	Metric	Pilot model (60 stocks)	This model, same 60 stocks	This model, all stocks
20-day volatility	R2	0.496	0.495	0.503
20-day volatility	wmR2	0.550	0.562	0.572
20-day volatility	MAE	0.268	0.265	0.272
drawdown of 10 % or more within 20 days	AUC	0.656	0.659	0.663
drawdown of 10 % or more within 20 days	BSS	0.062	0.077	0.089
drawdown of 10 % or more within 20 days	ECE	0.058	0.034	0.028
beats the median stock over 5 days	AUC	0.510	0.510	0.508

Limits

- 120 of 979 stocks; the full-universe run uses the same notebook with QUICK = False.
- Survivor-biased universe (only stocks that still exist) from one data vendor, so risk may be understated for stocks that later fail.
- The chart renderer scales each picture to fill its panel, so pictures cannot show the volatility level; numbers carry that.
- Live dashboard inputs come from Yahoo Finance, which can differ slightly from the training vendor.
- Research output, not investment advice. NO TRADE is the default decision.

Glossary

Term	Meaning
R^2	Share of the ups and downs in the target that the model explains. 1 = perfect, 0 = no better than always guessing the average.
AUC	Chance that the model ranks a case where the event happened above one where it did not. 0.5 = coin flip, 1 = perfect.
Brier skill score	How much better the probabilities are than always predicting the average rate (0 = no better).
Calibration	Whether stated chances come true as often as stated (30% events happen about 30% of the time).
Embedding / PCA	The numbers a vision network outputs for a picture / a way to compress them to the most informative 128.
Month-block bootstrap	Confidence intervals made by resampling whole calendar months, because stocks move together within a month.
Holdout	The most recent data (2023 onward), sealed until all choices are frozen and then opened once.